DS	**Session**	Stack	**Question Information**	Level 1 Challenge 41

Problem Description

A and B are playing a game. In this game, both of them are initially provided with a **list of n numbers**. (Both have the same list but their own copy).

Now, they both have a different strategy to play the game. A picks the element from **start of his list**. B picks from the **end of his list**.

You need to generate the result in form of an output list.

Method to be followed at each step to build the output list is:

1. If the number picked by A **is bigger than** B then this step's **output is** 1 . B **removes** the number that was picked from their list.
2. If the number picked by A **is smaller than** B then this step's **output is** 2 . A **removes** the number that was picked from their list.
3. If both have the **same number** then this step's **output is** 0 . **Both** A **and** B **remove** the number that was picked from their list.

This game **ends** when at least one of them has no more elements to be picked i.e. when the **list gets empty**.

Output the built output list.

Constraints

1 ≤ N ≤ 10^6

1 ≤ num ≤10^9

Input format:

First line consists of a number n , size of the list provided.
Next line consists of n numbers separated by space.

Output format:

Output the required output list.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3
1 2 3
```

EXPECTED OUTPUT

```
2 2 0
```

Test Case 2

INPUT (STDIN)

```
5
5 7 1 2 3
```

EXPECTED OUTPUT

```
1 1 1 2 0 2 2 2
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
if(a[i]>a[j])
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
6
```

Test Case 2

TOKEN COUNT

```
160
```

Test Case 3

NLOC

```
36
```

Code

solution.c

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int a[n];
    for (int i = 0; i < n; i++) scanf("%d", &a[i]);
    int i = 0, j = n - 1;
    while (i < n && j >= 0) {
        if (a[i] > a[j]) {
            printf("1 ");
            j--;
        } else if (a[i] < a[j]) {
            printf("2 ");
            i++;
        } else {
            printf("0 ");
            i++;
            j--;
        }
    }
    printf("\n");
    return 0;
}
```

Output

Problems Output Debug Console Terminal Ports powershell

```
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
3
1 2 3
2 2 0
PS C:\C Programming>
```